

Long-Term Personalized Adaptive Deep Brain Stimulation in Parkinson Disease

A Nonrandomized Clinical Trial

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 Supplemental content

IMPORTANCE Adaptive deep brain stimulation (aDBS) automatically adjusts stimulation amplitude in response to changes in relevant neural activity in people with Parkinson disease (PD). Whether long-term at-home aDBS is safe and delivers effective therapy in people with PD remains unknown.

OBJECTIVE To determine the tolerability, efficacy, and safety of long-term aDBS in people with PD who were previously stable receiving continuous DBS (cDBS).

DESIGN, SETTING, AND PARTICIPANTS This international, open-label, prospective, pivotal trial enrolled participants from December 2020 to July 2022 in the US, Canada, and Europe. Referred participants with PD were first assessed while receiving stable cDBS and those who tolerated 2 aDBS modes were randomized and blinded to 30 days in each mode (single-blind crossover design); those who tolerated only 1 mode were assessed in that mode only; assessments completed holding medication stable. Participants were given the option to continue their selected mode of aDBS for long-term follow-up (10 months). Data used for analysis were from March 2024. Multiple imputation was used if more than 5% of data was missing for the primary or secondary end points. A referred sample of 68 participants with PD, stable while receiving cDBS and medication, was included.

INTERVENTIONS Two modes of aDBS controlled by an embedded closed-loop stimulation system: single threshold (ST-aDBS) and dual threshold (DT-aDBS).

MAIN OUTCOMES AND MEASURES The primary end point required that at least 50% of participants meet a performance goal of on-time (ie, time when symptoms were well controlled) without troublesome dyskinesias with no less than 1-SD reduction (and post hoc threshold less than 2 hours per day reduction) reported during aDBS therapy compared to cDBS, determined from a self-reported motor diary. The secondary end point was total electrical energy delivered (TEED) compared between aDBS and cDBS. Safety assessments were conducted by characterizing adverse events (AEs), stimulation-related AEs, serious AEs, and device deficiencies.

RESULTS A total of 68 participants enrolled (mean [SD] age, 62.2 [8.4] years; 48 [70.6%] male); 40 and 35 were evaluated with DT-aDBS and ST-aDBS, respectively. The primary end point performance goal was met in the DT-aDBS group (91% of participants) and ST-aDBS (79% of participants) with the post hoc performance threshold; no difference between aDBS modes ($\chi^2_1 = 1.0$; $P = .51$). TEED was reduced during ST-aDBS compared to cDBS (mean change, -15%; nominal $P = .01$) and not different from DT-aDBS. All but 1 stimulation-related AE resolved during the aDBS setup and adjustment phase with no serious device AEs through long-term follow-up. Exploratory analyses suggested improvement in on-time without troublesome dyskinesias with DT-aDBS compared to cDBS.

CONCLUSIONS AND RELEVANCE In this study, long-term aDBS was tolerable, effective, and safe in people with PD who were previously stable while receiving cDBS.

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Continuous deep brain stimulation (cDBS) is standard of care in Parkinson disease (PD),^{1,2} with combined cDBS and medication being superior to medication alone for the treatment of PD.^{3,4} Currently available DBS devices deliver continuous neurostimulation at 1 constant amplitude and cannot respond to dynamic fluctuations of PD symptoms or natural or medication-induced fluctuations in underlying neural activity. For some people with PD receiving combined cDBS plus medication, this results in overtreatment when medications are at their peak dose and undertreatment when medications are wearing off.

Adaptive deep brain stimulation (aDBS) automatically adjusts DBS parameters in response to fluctuations in neural activity or behavior to provide more stable symptom control, reduce adverse effects, and potentially increase neurostimulator battery longevity.⁵⁻⁹ Subthalamic nucleus (STN) local field potential (LFP) power and burst duration in the β (± 13 -30 Hz) and α (8-12 Hz) band are promising control signals for aDBS. They are attenuated by both dopaminergic medication and DBS,¹⁰⁻¹⁵ proportional to the degree of improvement in motor symptoms, and linked to symptom severity in the off state.^{10-12,16-23}

We report the results of the Adaptive DBS Algorithm for Personalized Therapy in Parkinson Disease (ADAPT-PD) pivotal trial, which investigated the tolerability, efficacy, and safety of long-term at-home aDBS driven by LFP power from frequencies within the α - β band (8-30 Hz) in people with PD who were previously stable receiving STN or globus pallidus internus (GPi) cDBS and medication. At the start of the ADAPT-PD trial, small, single-center studies had demonstrated feasibility and efficacy of short periods of aDBS under clinical observation⁶⁻⁸; however, it was not known whether long-term at-home aDBS would be tolerable, efficacious, and safe. Since no at-home aDBS data existed at the time of trial design, the objective was to demonstrate the at-home tolerability, efficacy, and safety of 2 previously reported aDBS modes: single threshold (ST-aDBS) and dual threshold (DT-aDBS).^{5,7,8,24} The primary objective was to demonstrate that at least 50% of participants would meet a performance threshold based on the change in on-time (ie, time when symptoms were well controlled) without troublesome dyskinesia reported while receiving aDBS vs cDBS, both stable with medication. The secondary objective was to assess the energy efficiency (total electrical energy delivered [TEED]) of the 2 aDBS modes in comparison to cDBS. The overarching purpose of ADAPT-PD was to determine if similar therapeutic performance could be achieved with long-term at-home aDBS compared to standard of care cDBS in people with PD.

Methods

Trial Design and Oversight

Clinical oversight for each site was provided by the principal investigator, and study oversight was provided by an independent data monitoring committee. The protocol was approved by the institutional review board or ethics committee at each participating institution with written informed consent provided by all participants, and the

Key Points

Question Is long-term adaptive deep brain stimulation (aDBS) tolerable and as effective and safe as continuous DBS (cDBS)?

Findings In this nonrandomized clinical trial with an open-label comparison between cDBS and aDBS, the primary outcome was met as the majority of participants receiving aDBS achieved a performance goal of good on-time (ie, time when symptoms were well controlled) without troublesome dyskinesia relative to stable cDBS therapy.

Meaning Long-term aDBS provided tolerable, effective, and safe therapy in persons with Parkinson disease whose symptoms were previously stable while receiving cDBS.

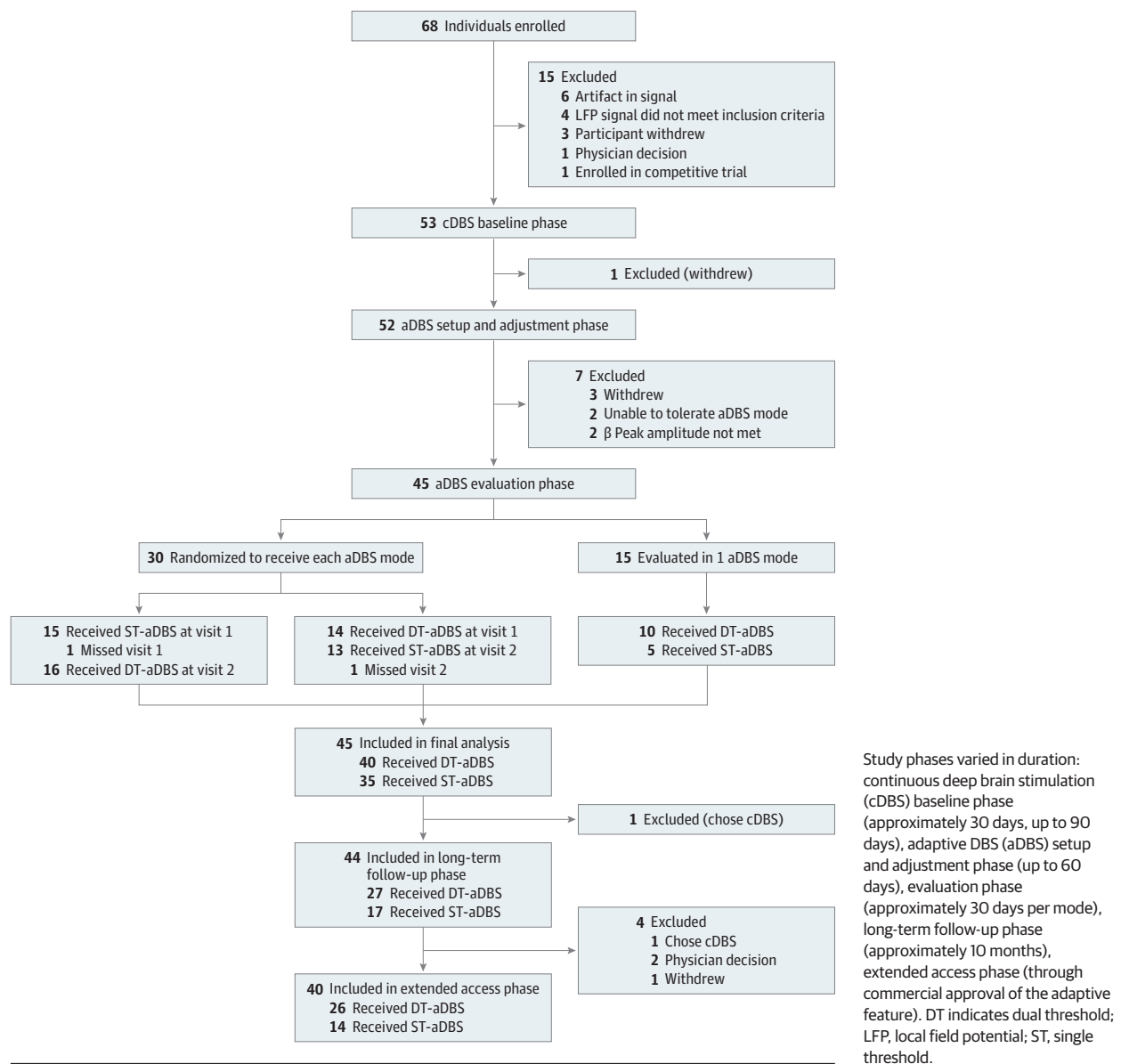
study was conducted in accordance with the Declaration of Helsinki and the Good Clinical Practice guidelines of the International Council for Harmonization. ADAPT-PD followed ISO14155 reporting guidelines.

ADAPT-PD was an international, multicenter, prospective, pivotal trial, with a randomized single-blind crossover (between 2 aDBS modes) trial to evaluate the tolerability, efficacy, and safety of aDBS in people with PD who were stable while receiving cDBS and medication. The primary end point required that at least 50% of participants would report on-time during aDBS within a performance goal of no more than -1 SD change compared to cDBS while maintaining stable medication regimens. The primary end point performance goal was revised post hoc to a time-based threshold of -2 hours per day loss of on-time without troublesome dyskinesias to ameliorate type I error risk (see the eMethods in Supplement 2 for a detailed description). TEED was chosen as a secondary outcome to investigate if stimulation energy could be reduced during aDBS and estimate the potential impact aDBS had on battery longevity compared to that of cDBS. When ADAPT-PD was designed (in 2018), it was unknown how aDBS would adjust stimulation when used in a patient's home or during sleep as all previous studies were limited to in-clinic assessments.^{5,7,25} There were also no sensing-enabled rechargeable implantable pulse generators commercially available, so it was important to estimate longevity impacts of aDBS in the event that aDBS patients would require battery replacement more often.

Participants

The study population included people with PD aged 18 years and older, previously implanted with bilateral DBS leads (Medtronic, Legacy [3387/3389, nonsegmented electrodes] or SenSight [segmented electrodes]) in the STN or GPi, which were connected to a commercial sensing-enabled DBS system (Medtronic, Percept PC).²⁶ Participants were required to be responsive to cDBS, be programmed in a sensing-compatible monopolar or dual-monopolar electrode configuration, have stable DBS parameters and medication regimens, and have an LFP peak amplitude of $1.2 \mu\text{V}$ peak or greater in the α - β band on left or right DBS leads (see eTable 1 in Supplement 2 for complete inclusion and exclusion criteria).

Figure 1. CONSORT Diagram of Study



Trial Procedures

The protocol has been previously published²⁷ and is available in [Supplement 1](#). The study consisted of 4 phases following enrollment with optional extended access (**Figure 1**; eFigure 1 in [Supplement 2](#)). The cDBS baseline phase (up to 90 days) included LFP screening (see section 9.8.1 of the study protocol in [Supplement 1](#)) and baseline cDBS assessments with the device both on and off. Participants then entered the aDBS setup and adjustment phase (up to 60 days), where the investigational aDBS firmware was unlocked to allow programming of the 2 different aDBS modes—ST-aDBS and DT-aDBS²⁷ (eFigure 2 in [Supplement 2](#)). Participants' devices were programmed with each mode of aDBS using the same frequency of interest, while not taking medication and then taking medication (conducted over 1 to 2 days). They were then sent home for a trial period within each mode of aDBS, with

additional adjustments allowed until either or both ST-aDBS and DT-aDBS modes were optimized. Participants were blind to the aDBS mode being tested. The aDBS setup and adjustment phase was followed by the evaluation phase: participants who tolerated both aDBS modes were randomized and blinded to 30 days of long-term ST-aDBS or DT-aDBS followed by 30 days in the other aDBS mode. No aDBS parameter changes were allowed within 14 days prior to the evaluation visit, which was completed at the end of each 30-day period. Changes in PD medications from enrollment until the end of the aDBS evaluation phase were only permitted if deemed absolutely necessary by the clinician. A single treatment evaluation phase was completed for participants who only tolerated 1 aDBS mode. At the end of the evaluation phase, participants who tolerated both aDBS modes were asked to choose in a blinded manner which mode (cDBS, ST-aDBS, DT-aDBS)

Table 1. Baseline Characteristics of Study Population

Characteristic	Primary cohort consented (N = 68)
Age, mean (SD) [range], y (n = 66)	62.2 ± 8.4 [36-75]
PD duration, mean (SD), y (n = 64)	13.5 (6.8)
Dyskinesia, mean (SD), y (n = 37)	6.9 (4.8)
Motor fluctuations, mean (SD), y (n = 46)	7.6 (4.6)
Duration of levodopa treatment, mean (SD), y (n = 60)	10.7 (6.1)
Levodopa equivalent daily dose, mean (SD), mg	561.9 (568.3)
Sex, No. (%)	
Male	48 (70.6)
Female	20 (29.4)
Target site by participant, No. (%)	
STN	51 (75.0)
GPI	17 (25.0)
Years from the lead implant to consent, mean (SD) [range]	3.4 (3.3) [0.2-11.3]
MDS-UPDRS part III, mean (SD) (n = 58) ^a	45.7 (14.9)
Tremor	8.8 (6.4)
Rigidity	8.3 (3.6)
Bradykinesia	22.9 (8.3)
Axial	5.6 (3.0)

Abbreviations: GPI, globus pallidus internus; MDS-UPDRS, Movement Disorders Society–Unified Parkinson's Disease Rating Scale; PD, Parkinson disease; STN, subthalamic nucleus.

^a Data collected with no current stimulation or medication.

they preferred and could continue receiving aDBS in their preferred mode (selected aDBS mode) during the long-term follow-up phase (approximately 10 months), during which there were 4 evaluations. At the completion of the long-term follow-up phase, participants had the option to further continue with aDBS in an extended access phase until commercial approval of aDBS. Details of the aDBS setup and choice of the personalized LFP band (neural physiologic marker) to drive aDBS have been previously reported.²⁷

Outcomes

The primary outcome of on-time without troublesome dyskinesias was based on a self-reported PD home diary²⁸ completed by participants every 30 minutes over 24 hours on at least 3 consecutive days within 14 days prior to the evaluation visit. Participants documented time as on-time without dyskinesia, on-time with nontroublesome dyskinesia (grouped together as on-time), on-time with troublesome dyskinesia, off time, and asleep time. The secondary outcome was TEED,²⁹ calculated during each aDBS evaluation phase compared to during cDBS. The tolerability of aDBS was assessed by the clinician using the aDBS Global Impression of Change assessment (eMethods in Supplement 2). An acceptable and tolerable response was defined as having no worse efficacy to cDBS and either no adverse effects or adverse effects that did not significantly interfere with the participant's functioning. An unacceptable and intolerable response was one that had either worse efficacy or adverse effects that significantly interfered with the participant's functioning. Safety was assessed by the frequency and severity of reported adverse events (AEs) throughout the study. Additional outcomes included differ-

ences in the Movement Disorders Society–Unified Parkinson's Disease Rating Scale (MDS-UPDRS, parts I, II, III, IV)³⁰ and assessments of quality of life and patient preference and satisfaction. MDS-UPDRS scores were collected without stimulation or medication at enrollment and with stimulation and medication at cDBS and aDBS evaluation. Other outcomes collected but not included in this report included wearable device data, sleep, and speech assessments.

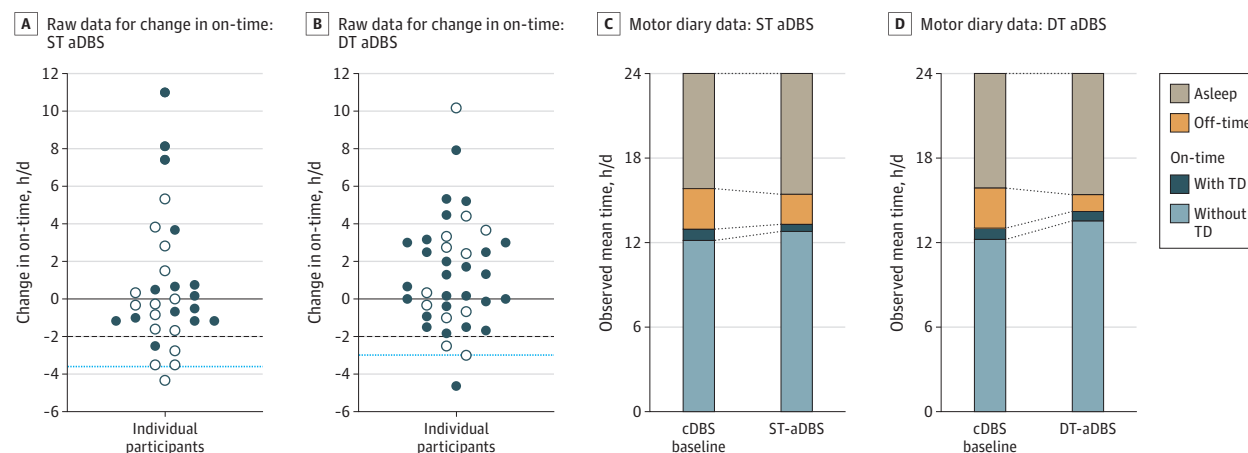
Statistical Analysis

Details of the statistical methods are provided in the statistical analysis plan (Supplement 1). The initial protocol defined the primary end point to be met if at least 50% of participants met a prespecified patient-specific performance threshold that was intended to estimate comparable on-time without troublesome dyskinesias in either mode of aDBS compared to cDBS. The patient-specific performance goal, defined as the change in daily on-time without troublesome dyskinesias reported during aDBS compared to cDBS, was computed for each participant. If the difference (aDBS minus cDBS) for each participant was greater than −1 SD, the aDBS performance threshold was met for the participant. The percentage of participants with on-time during the respective aDBS evaluation phase that exceeded this threshold was compared to 50% using a binomial exact test for each aDBS mode. Multiple imputation was used if more than 5% of data was missing for the primary or secondary end points.

Post hoc, it was determined that the initial performance threshold and planned analyses violated 2 key assumptions of the exact binomial test: the success threshold depended on the sample standard deviation and the distribution of the ST-aDBS change of on-time was not normally distributed. These violations had the potential to inflate the risk of type I error and therefore the performance threshold was revised post hoc. The revised performance threshold and end point required that at least 50% of participants achieved the performance outcome where the reduction in on-time without troublesome dyskinesias reported during DBS was no more than 2 hours per day when using aDBS compared to their stable cDBS. This performance threshold is more conservative than the originally planned analysis method and ameliorates type I error concerns. Since the primary objective was evaluated using a post hoc analysis, confirmatory inference through hypothesis testing is precluded. Hence, the *P* values presented here are nominal and are provided for descriptive purposes. The rationale for the 50% performance threshold (50% of patients with comparable on-time) was chosen to demonstrate that the majority of patients had comparable efficacy during the day with aDBS compared to stable cDBS. Additional post hoc analyses were completed to evaluate patients receiving their selected mode of aDBS chosen at long-term follow-up with raw change scores and effect sizes. Data were summarized with standard descriptive summary statistics.

The sample size was calculated using a binomial distribution for a 1-sided exact test ($\alpha = .0125$), alternative hypothesis of 85%, resulting in a minimum of 36 participants to achieve at least 90% power to reject a performance goal of 50%. ST-aDBS and DT-aDBS modes were evaluated separately using

Figure 2. Changes in Motor Diary



A and B show the raw data for change in on-time (ie, time when symptoms were well controlled) without troublesome dyskinesia for each adaptive deep brain stimulation (aDBS) mode assessed compared to continuous DBS (cDBS) evaluation during single-threshold (ST) and dual-threshold (DT) modes. Filled datapoints indicate if the participant chose that mode at the long-term follow-up, unfilled dots indicate if the patient chose a different mode at the long-term follow-up. The prespecified performance threshold (-1 SD, dotted line) and post hoc threshold (-2 hours per day, dashed line) were used to determine success or failure to calculate the primary end point. C and D show the motor diary data evaluated during ST and DT as observed mean hours per day spent asleep, off-time (ie, time when symptoms were poorly controlled), and on-time with and without troublesome dyskinesia.

a Bonferroni-corrected α ($.025 / 2 = .0125$ for the P value cut-off) for the respective hypothesis.

Results

Participant Demographic Characteristics

A total of 68 people (mean [SD] age 62.2 [8.4] years; 48 [70.6%] male) with moderate to advanced PD,^{3,4} who were receiving stable cDBS and medication, were enrolled between December 14, 2020, and April 29, 2022 (Table 1, Figure 1; eTable 2 in Supplement 2). The majority of screen failures occurred at enrollment due to the presence of signal artifact or insufficient amplitude of the α - β peak²⁷ (eTable 3 in Supplement 2). LFP peak detection was successful in 44 of 54 legacy leads (81.5%) and 13 of 14 SenSight leads (92.9%) at the LFP screening visit. The peak frequency of the band selected to drive aDBS spanned 9 to 29 Hz (eFigure 3 in Supplement 2), with 3 participants using a peak less than 10 Hz. The median peak frequency used for aDBS was 17.6 Hz and 18.6 Hz on the left and right hemispheres, respectively, which were not significantly different (mean [SD] difference, -1.03 [4.98]; signed rank test $P = .37$; $n = 45$). Programming parameters are provided in eTable 4 in Supplement 2.

Tolerability and Participant Choice to Continue aDBS

Of the 52 participants who entered the aDBS setup and adjustment phase, 45 (87%) tolerated at least 1 mode and 30 (58%) participants tolerated both modes. Of the 15 participants who tolerated only 1 mode, 5 were assessed who were receiving ST-aDBS and 10 receiving DT-aDBS. After the evaluation phase, 44 of 45 participants chose to continue aDBS and entered the long-term follow-up phase. During the long-term follow-up phase, 4 participants exited the study due to preference for cDBS ($n = 1$),

physician decision ($n = 2$), or difficulty traveling to the study site ($n = 1$). Forty participants continued with aDBS in the extended access phase. Levodopa equivalent doses were similar on aDBS compared to cDBS (eTable 5 of Supplement 2).

Primary End Point

The prespecified primary outcome (-1 SD performance threshold) was met: 95% of participants receiving DT-aDBS (exact 97.5% lower confidence limit, 81.2%; $n = 40$) and 87% of participants receiving ST-aDBS (exact 97.5% lower confidence limit, 68.9%; $n = 35$) had comparable on-time to cDBS (Figure 2A and B; Table 2; sensitivity analysis in eTable 6 of Supplement 2). The post hoc analysis (-2 hour per day performance threshold) primary end point was also met: 91% of participants receiving DT-aDBS (exact 97.5% lower confidence limit, 75.6%; $n = 40$) and 79% of participants receiving ST-aDBS (exact 97.5% lower confidence limit, 59.4%; $n = 35$) had comparable on-time to cDBS (both $P < .001$). Individual participant data are provided in Figure 2A and B with the performance thresholds to determine the percentage of participants passing the primary end point as the horizontal lines. The number of participants meeting the primary end point did not differ between aDBS modes (exact McNemar test, 17 concordant pairs; $\chi^2_1 = 1.0$; $P = .51$).

Secondary End Point

The secondary end point was met for ST-aDBS, in which TEED was lower compared to cDBS (change during ST-aDBS compared to cDBS, -22.3 μ Watts; 97.5% CI, -42.0 to -2.6 ; $P = .01$) (Table 2). DT-aDBS TEED was also lower (change, -22.3 μ Watts; 97.5% CI, -48.0 to -3.3 ; $P = .05$). TEED during aDBS was not different between ST-aDBS and DT-aDBS modes ($F_{26} = 0.30$; $P = .59$, eTable 7 in Supplement 2). In the participant-selected aDBS mode, there was lower TEED compared

Table 2. Primary and Secondary End Points, Adaptive Deep Brain Stimulation (aDBS) vs Continuous DBS (cDBS) by Mode (Full Analysis Set)

Categorical end point	Mode	Threshold	Proportion of success ^a	Exact 97.5% lower confidence limit ^a	Nominal <i>P</i> value ^b	
On-time without troublesome dyskinesia ^c	Single (n = 35)	−1 SD ^d	87%	68.9	NA	
	Dual (n = 40)	−1 SD ^d	95%	81.2	NA	
	Single (n = 35)	−2 h/d	79%	59.4	<.001	
	Dual (n = 40)	−2 h/d	91%	75.6	<.001	
Continuous end point	Mode	cDBS, mean (SE)	aDBS, mean (SE)	Mean difference (97.5% CI)	Effect size (mean % change)	Nominal <i>P</i> value ^e
TEED (μWatts)	Single (n = 35)	148.1 (16.77)	125.0 (14.11)	−22.3 (−42.0 to −2.6)	0.45 (−15)	.01
	Dual (n = 40)	172.1 (22.40)	148.5 (17.34)	−22.3 (−48.0 to −3.3)	0.32 (−13)	.05
	Selected (n = 45)	166.3 (20.67)	137.4 (15.63)	−27.0 (−48.9 to −5.1)	0.43 (−16)	.007

^a Multiple imputation Clopper-Pearson mean success approach.^d Original protocol-based performance threshold. No P value provided.^b Median of the P values from the multiple imputation datasets; statistically significant if 1-sided P < .0125.^e Multiple imputation statistical analyses were performed assuming normality or nonnormality depending on the distribution of the change variable.^c Defined as time when symptoms were well controlled.

to cDBS (-27.0 μWatts; 97.5% CI, -48.9 to -5.1; P = .007). A sensitivity analysis demonstrated reduced TEED in both modes for the complete case dataset and the as-treated dataset (eTable 8 in Supplement 2).

Safety

Stimulation-related AEs occurred during the aDBS setup and adjustment phase, with events largely categorized as worsening of PD symptoms (22% of events) as would be expected when optimizing DBS settings (Table 3). All but 1 stimulation-related AEs resolved with reprogramming during the aDBS setup and adjustment phase (59 of 60 events resolved) (Table 3) and all stimulation-related events resolved during the aDBS evaluation phase (17 of 17 events resolved) (Table 3). The 1 unresolved stimulation-related event was insomnia, which was deemed to not significantly interfere with patient's functioning. It was expected that stimulation-related AEs would be resolved during aDBS setup, as tolerability was necessary for participant entry to the evaluation phase. There were no differences between aDBS modes in the proportion of participants with stimulation-related AEs during the aDBS evaluation phase (eTable 9 in Supplement 2). No unanticipated serious adverse device events were reported from enrollment to the end of the long-term follow-up phase. The average exposure to aDBS from the start of the aDBS evaluation phase to last follow-up of this reported data was 17.5 months with a range of 3 to 27 months.

Participant Choice of aDBS vs cDBS

Among the 45 participants completing aDBS evaluation, 27 (60%) selected DT-aDBS and 17 (37.8%) selected ST-aDBS for long-term follow-up. One participant (2.2%) selected cDBS and exited the study after the evaluation phase. Participants reported preferring aDBS over cDBS for improvement of motor symptoms (63% of participants), or reduction of adverse effects (45% of participants) and symptom fluctuations (66% of participants) (eTable 10 in Supplement 2).

Additional Exploratory Clinical Outcomes

On-time increased and off time decreased with DT-aDBS compared to cDBS (difference, 1.3 hours per day and -1.6 hours per day, respectively; effect size, 0.41-0.63) (eTable 11 in Supplement 2; Figure 2B and D). On and off time were similar with ST-aDBS compared to cDBS. On-time with aDBS was not different between ST-aDBS and DT-aDBS (Figure 2A and C; eTable 12 in Supplement 2). In a post hoc comparison using the mode that was selected for the long-term follow-up phase, on-time increased and off time decreased compared to cDBS (difference, 1.4 hours per day and -1.7 hours per day, respectively; effect size, 0.42-0.56) (eTable 11 in Supplement 2). Time spent asleep increased during selected mode aDBS compared to cDBS (difference, 0.5 hours per day; effect size, 0.37) (eTable 11 in Supplement 2).

MDS-UPDRS III score was lower during DT-aDBS compared to cDBS (difference, -2.6; effect size 0.35) (eTable 13 in Supplement 2), lower during DT-aDBS and ST-aDBS compared to off stimulation/off medication (effect size, 1.76-2.00) (eTable 13 in Supplement 2), and not different between ST-aDBS and DT-aDBS (eTable 13 in Supplement 2). Post hoc comparisons indicated that both the MDS-UPDRS parts III³¹ and IV³² scores were lower with selected mode aDBS than on cDBS (effect size, 0.48) (eTable 13 and 14 in Supplement 2). Quality of life as assessed by the Parkinson's Disease Questionnaire (PDQ-39) summary index was not different with any aDBS mode compared to cDBS (eTable 11 in Supplement 2).

Discussion

ADAPT-PD was an international, multicenter, pivotal trial to assess the tolerability, efficacy, and safety of long-term at-home aDBS and medication compared to standard of care (ie, stable cDBS and medication). The study met its primary and secondary end points. The majority of participants receiving aDBS experienced comparable on-time without troublesome dyskinesia compared to stable cDBS. TEED was lower during ST-aDBS

Table 3. Frequency of Stimulation-Related Adverse Events by System Organ Class, Preferred Term, and Study Phase (Primary Cohort)^a

System organ class	No. of events (No. of participants) [% of participants]			
	Enrollment through cDBS baseline phase (1 mo) (n = 68)	aDBS setup and adjustment phase (1-2 mo) (n = 53)	aDBS evaluation phase (1-2 mo) (n = 45)	Long-term follow-up phase (10 mo) (n = 44)
All adverse events	3 (3) [4.4%]	60 (30) [56.6%]	17 (13) [28.9%]	16 (13) [29.5%]
Psychiatric disorders				
Insomnia	0	2 (2) [3.8%]	0	0
Apathy	0	1 (1) [1.9%]	0	0
Depressed mood	0	0	0	1 (1) [2.3%]
Impulse-control disorder	1 (1) [1.5%]	0	0	0
Nervous system disorders				
Parkinson disease symptoms (worsening)	0	13 (12) [22.6%]	6 (6) [13.3%]	4 (4) [9.1%]
Dyskinesia	0	17 (13) [24.5%]	2 (1) [2.2%]	3 (2) [4.5%]
Tremor	1 (1) [1.5%]	5 (4) [7.5%]	2 (2) [4.4%]	0
Dystonia	0	2 (1) [1.9%]	0	1 (1) [2.3%]
Paresthesia	0	4 (3) [5.7%]	1 (1) [2.2%]	0
Freezing phenomenon (freezing of gait)	1 (1) [1.5%]	0	1 (1) [2.2%]	2 (2) [4.5%]
Parkinsonian rest tremor	0	3 (3) [5.7%]	0	1 (1) [2.3%]
Dizziness	0	3 (3) [5.7%]	0	0
On and off phenomenon	0	1 (1) [1.9%]	0	1 (1) [2.3%]
Bradykinesia	0	1 (1) [1.9%]	0	0
Dysarthria	0	1 (1) [1.9%]	0	0
Hemiparesthesia	0	1 (1) [1.9%]	0	0
Muscle contractions involuntary	0	1 (1) [1.9%]	0	0
Eye disorders				
Blepharospasm	0	1 (1) [1.9%]	0	0
Vascular disorders				
Hot flush	0	1 (1) [1.9%]	0	0
Respiratory, thoracic, and mediastinal disorders				
Dysphonia	0	0	0	1 (1) [2.3%]
Gastrointestinal disorders				
Nausea	0	1 (1) [1.9%]	0	0
Musculoskeletal and connective tissue disorders				
Musculoskeletal stiffness	0	0	2 (2) [4.4%]	0
Muscle rigidity	0	1 (1) [1.9%]	0	0
Muscle twitching	0	0	0	1 (1) [2.3%]
Pain in jaw	0	0	1 (1) [2.2%]	0
General disorders and administration site conditions				
Asthenia	0	0	1 (1) [2.2%]	0
Sluggishness	0	1 (1) [1.9%]	0	0
Injury, poisoning, and procedural complications				
Fall	0	0	1 (1) [2.2%]	0
Product issues				
Device stimulation issue (dyskinesia due to overstimulation)	0	0	0	1 (1) [2.3%]

^a Adverse event classifications were adjudicated by the Adaptive DBS Algorithm for Personalized Therapy in Parkinson Disease (ADAPT-PD) Clinical Events Committee using the Medical Dictionary for Regulatory Activities (MedDRA). Parkinson disease (worsening) was used to categorize when more than 1 PD symptom was present.

compared to cDBS, and no serious device-related AEs occurred during the aDBS evaluation phase or long-term follow-up. There was no difference between ST-aDBS and DT-aDBS in either the primary or secondary outcome. To our knowledge, these re-

sults are the first to show that long-term at-home STN or GPi aDBS, driven by LFP power in the 8- to 30-Hz range and using algorithms embedded on an implanted system, was tolerable, safe, and effective relative to standard of care cDBS.²¹

Most participants (45 of 52 [87%]) found aDBS programming tolerable and entered the evaluation phase. Two participants did not tolerate aDBS and 15 participants tolerated 1 aDBS mode. Overall, the safety profile observed in this study for aDBS is consistent with those described in Medtronic's current labeling for DBS for PD.

All but 1 participant (44/45) chose to continue receiving aDBS when entering the 10-month follow-up phase, and 40 of 44 participants chose to continue receiving aDBS into the extended access phase (more than 1 year of aDBS). More than two-thirds of study participants indicated a "strong" or "somewhat" preference for aDBS over their previous cDBS settings for improved motor symptoms or fewer symptom fluctuations (eTable 10 in [Supplement 2](#)).

While TEED was lower on average during both aDBS modes compared to during cDBS, the impact of aDBS on battery life is more nuanced. Battery longevity is dependent on both TEED and the energy required to run the adaptive algorithm, such as sampling rate (20 Hz with ST-aDBS and 5 Hz with DT-aDBS) and the frequency of stimulation amplitude adjustments. Using the median change in longevity from the ADAPT-PD cohort (calculated from participant data with 7 days of stable aDBS prior to randomization), we estimate that battery longevity is improved by 5% per year for DT-aDBS and reduced by 4% per year for ST-aDBS compared to cDBS without sensing enabled. cDBS with passive sensing uses 3% per year; therefore, DT-aDBS would lead to greater battery longevity on average, and ST-aDBS would nearly break even compared to cDBS with passive sensing. Given the range of parameters used in practice, clinicians should weigh individual battery life impact (if using primary cell, Percept PC) against the potential therapeutic benefits experienced by each patient. This is less of a concern with the availability of a rechargeable sensing-enabled neurostimulator (Medtronic, Percept RC).

Exploratory clinical outcomes suggested improvements in on and off time (Figure 1; eTable 10 in [Supplement 2](#)), as well as in motor severity (eTable 13 in [Supplement 2](#)), with DT-aDBS and selected mode compared to stable cDBS. The on and off times recorded at baseline while receiving cDBS in the ADAPT-PD trial were similar to those reported while receiving cDBS in previous large multicenter trials^{3,4} (eFigure 4 in [Supplement 2](#)), indicating that the ADAPT-PD cohort entered the trial with similar cDBS efficacy to that reported in the multicenter trials, which were foundational in establishing the efficacy of cDBS therapy. Although improvement in clinical outcomes beyond baseline cDBS therapy were observed in our cohort, this result should be interpreted with caution, as participants and

investigators were not blinded with respect to the aDBS and cDBS assessments.

Over the course of the ADAPT-PD trial, single-center studies have reported the efficacy of aDBS outside of the clinic or in a home environment.³³⁻³⁵ aDBS was found to be superior to cDBS but with a higher TEED in a naturalistic environment in 4 people with PD implanted with an investigative neurostimulator using a blinded and randomized design.³³ This study used an alternative γ -band-based control algorithm, and it remains unknown if α - β and γ -controlled aDBS provide similar clinical benefits. Future studies will be needed to better understand what clinical benefits aDBS may offer to optimize aDBS programming workflows and to define optimal aDBS patient selection.

The enrolled participants represented a diverse population, including those with STN and GPi DBS, 2-lead constructions, a range of disease severities, and included both recent and long-standing DBS implants (0.2-11.3 years since lead implantation [Table 1]). The sample size is too small for a statistical comparison between the subgroups. Whether aDBS efficacy is comparable between de novo and long-standing DBS implants remains to be determined.

Limitations

Limitations of the ADAPT-PD study included a relatively small sample size evaluated while receiving aDBS. The sample size is insufficient to identify specific clinical or demographic features predicting aDBS outcomes. Importantly, comparisons of performance between aDBS and cDBS must be interpreted with caution, as participants and investigators were not blinded to aDBS state and placebo effects may inflate our results.^{32,33} However, the high rate of participants selecting to continue with aDBS over cDBS (44 of 45 [98%]) and the long duration of treatment are encouraging for future clinical use in PD. The field of closed-loop neuromodulation for PD is rapidly evolving and other compelling control signals and algorithms are likely to be discovered.

Conclusions

The ADAPT-PD trial demonstrated the tolerability, efficacy, and safety of long-term at-home aDBS in 45 people with moderate to advanced PD who were previously receiving stable cDBS. Delivering adaptive neurostimulation chronically, in response to a personalized neural physiologic marker, is both technically and clinically possible, laying the foundation for the wider clinical application of aDBS.

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